Q 1) a: STAT 4C03

$$L(\beta) = \prod_{i=1}^{n} \pi_i^{y_i} (1-\pi_i)^{1-y_i}$$

using the probit link, we have $\pi_i = F(x_i^t \beta) = F(\eta_i)$

F is the normal CFD. Let f be the normal density function

So the log-likelihood is:

$$\ell(\beta) = \sum_{i=1}^{n} y_i \log(\pi_i) + (1-y_i) \log(1-\pi_i) \quad \text{and}$$

we have:

for $j = 0, 1$ $\quad \frac{\partial \ell(\beta)}{\partial \beta_j} =$

$$\frac{\partial \ell(\beta)}{\partial \beta_j} = \sum_{i=1}^{n} y_i \frac{\partial}{\partial \beta_j} (\log(F(x_i^t \beta))) + (1-y_i)$$

$$+ (1-y_i) \frac{\partial}{\partial \beta_j} (\log(1-F(x_i^t \beta)))$$

$$= \sum_{i=1}^{n} y_i \frac{x_{ij} f(x_i^t \beta)}{F(x_i^t \beta)} - \frac{x_{ij} (1-y_i) f(x_i^t \beta)}{1-F(x_i^t \beta)}$$

$$\frac{\partial \ell(\beta)}{\partial \beta_j} = \sum_{i=1}^{n} x_{ij} f(x_i^t \beta) \left(\frac{y_i}{F(x_i^t \beta)} - \frac{(1-y_i)}{1-F(x_i^t \beta)} \right)$$

$$\boxed{\frac{\partial \ell(\beta)}{\partial \beta} = \sum_{i=1}^{n} x_{ij} f(\eta_i) \left(\frac{y_i - \pi_i}{\pi_i (1-\pi_i)} \right)}$$

Finally the equation for β_0 and β_1 are as follows :

We know that for $j = 0$, $x_{i0} = 1$ (intercept)

for β_0 :

$$\sum_{i=1}^{n} f(\eta_i)\left(\frac{y_i - \pi_i}{\pi_i(1-\pi_i)}\right) = 0 \quad (1)$$

for β_1 :

$$\sum_{i=1}^{n} x_{i1}\, f(x_i^t \beta)\left(\frac{y_i - \pi_i}{\pi_i(1-\pi_i)}\right) = 0 \quad (2)$$

b)

The matrix of second derivatives of l is given by $J_{kj} = \dfrac{\partial^2 l(\beta)}{\partial \beta_k \beta_j} = \dfrac{\partial}{\partial \beta_k}\left(\dfrac{\partial}{\partial \beta_j} l(\beta)\right)$

- We have $\dfrac{\partial}{\partial \beta_k}\left[\dfrac{y_i - F(\eta_i)}{F(\eta_i)(1-F(\eta_i))}\right] = -x_{ik}\left[\dfrac{y_i f(\eta_i)}{F(\eta_i)^2} + \dfrac{(1-y_i) f(\eta_i)}{(1-F(\eta_i))^2}\right]$

- and $\dfrac{\partial f(\eta_i)}{\partial \beta_k} = -x_{ik}\, \eta_i\, f(\eta_i)$ by the chain rule

by using the derivative rule $(u \cdot v)' = u'v + uv'$ and factorizing the common x_{ik} terms we get :

$\forall j,k \in \{0,1\}$,

$$J_{kj} = -\sum_{i=1}^{n} x_{ij} x_{ik} f(\eta_i) \left[y_i \left(\frac{f(\eta_i) + \eta_i F(\eta_i)}{(F(\eta_i))^2} \right) + (1-y_i) \left(\frac{f(\eta_i) - \eta_i (1 - F(\eta_i))}{(1 - F(\eta_i))^2} \right) \right]$$

$$J_{kj} = -\sum_{i=1}^{n} x_{ij} x_{ik} f(\eta_i) \left[y_i \left(\frac{f(\eta_i) + \eta_i \bar{\pi}_i}{\bar{\pi}_i^2} \right) + (1-y_i) \left(\frac{f(\eta_i) - \eta_i (1 - \bar{\pi}_i)}{(1 - \bar{\pi}_i)^2} \right) \right]$$

c) We have $\beta_1 = 0$ then $x_i^t \beta = \beta_0$, using (1) we get :

$$\sum_{i=1}^{n} f(\beta_0) \left(\frac{y_i - F(\beta_0)}{F(\beta_0)(1 - F(\beta_0))} \right) = 0$$

$$\iff F(\beta_0) = \frac{1}{n} \sum_{i=1}^{n} y_i = \bar{y} \text{ and finally :}$$

$$\hat{\beta}_0 = F^{-1}(\bar{y})$$

, F^{-1} : normal quantile function